

# Two gene-set tests — and why a robust claim needs both

Every program is run through two tests that ask different questions. Each, alone, has a blind spot.

SELF-CONTAINED · ROAST

Do the program’s own genes actually change — up or down — at all?

tests the set’s fold-changes against “no change” (zero); ignores other genes.

✓ immune to the relative-share effect — what other genes do can’t fake it.

✗ alone: if the whole genome drifts, the set can look like it moves too.

COMPETITIVE · CAMERA · GSEA

Does the program move MORE than the rest of the genome?

compares the set’s shift against the background of all other genes.

✓ shows the change is specific to this program, not everywhere.

✗ alone: a few strong genes can make the whole set look significant, even if most of it doesn’t move.

|               | SELF-CONTAINED moves ✓                                                | SELF-CONTAINED flat ✗                                                                                  |
|---------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| COMPETITIVE ✓ | <b>ROBUST — real &amp; specific ✓ keep</b><br>→ pericentral detox     | <b>Not coherent — not robust</b><br>⇒ PC identity phase II (relative-share effect) — not the whole set |
| COMPETITIVE ✗ | <b>Genome-wide drift</b><br>not specific no more than everything else | No change                                                                                              |

**A robust positive claim needs BOTH ticks:** self-contained says it’s real (its own genes moved off zero); competitive says it’s specific (more than the background). Only pericentral detox passes both — that’s our one confident finding.

# The stats terms in your part, in plain words

## $\rho$ · Spearman

How strongly two things move together, by rank, from  $-1$  to  $+1$ .  $\rho = -0.48 \rightarrow$  a moderate trend: detox falls as fibrosis rises. ( $-1$  = perfect opposite ·  $0$  = no link ·  $+1$  = perfectly together.)

## p-value

The chance of seeing a trend this strong if there were really nothing there — pure noise.  $p = 0.003 \rightarrow$  about 3 in 1,000  $\rightarrow$  very unlikely a fluke. The usual bar is  $p < 0.05$ .

## FDR · q-value

A p-value corrected for testing thousands of genes at once.  $FDR = 0.05 \rightarrow$  about 5% of the called “hits” are expected to be false. This is why a single gene must clear FDR, not raw p.

## NES · GSEA

How strongly a gene set piles up at the down (or up) end of the genome-wide ranking, vs random. More negative = a stronger coordinated down-shift of the whole set.